Chapter 7

Sensors

Robots are systems that sense, actuate, and compute. So far, we have studied the basic physical principles of motion, i.e., locomotion and manipulation. We now need to understand the basic principles of robotic sensors that provide the necessary data for a robot to make decisions and control itself. The goals of this chapter are to:

- provide an overview of sensors commonly used on robotic systems,
- outline the physical principles behind the functioning of sensors, and
- clarify the mechanisms responsible for uncertainty in sensor-based reasoning.

Historically, the development of robotic sensors is driven by industries other than robotics. These include transportation (automotive, naval, airplanes), safety devices for industry, servos for remote-controlled toys, and more recently cellphones, virtual reality, and gaming consoles. These industries are mostly responsible for making "exotic" sensors available at low cost by identifying mass-market applications. For example, accelerometers and gyroscopes are now widely used in smartphones and cost less than a dollar; the XBox gaming console made 3D depth sensing (through the Kinect) accessible for a greatly lower cost than before; and sensors in modern passenger vehicles provide an array of capability without appreciably increasing the cost of the vehicle itself.

Think about the sensors that you are interacting with daily. What sensors do you have in your phone, in your kitchen, or in your car?

As we will see later on, sensors are hard to classify by their application domain and target use case. In fact, most problems benefit from every possible source of information they can obtain. For example, localization can be achieved by measuring how many degrees a wheel has turned with a sensor known as an "encoder" (7.2), a sensor that can be implemented relying on a wide variety of physical principles. However, estimation becomes more precise with the addition of accelerometers (Section 7.3) or even vision sensors (Chapter 8). All of these approaches differ drastically in their precision—a term that will be more formally introduced below—and the kind of data they provide, but none of them is able to completely solve the localization problem on its own.

Think about the kind of data that you can obtain from an encoder, an accelerometer, or a vision sensor on a non-holonomic robot. What are the fundamental differences? What physical principles do they leverage?

Although an encoder is able to measure position, it is used in this function only on robotic arms. If the robot is non-holonomic, closed paths in its configuration space (i.e., robot motions that return the encoder values to their initial position), do not necessarily drive the robot back to its starting point (as exemplified in Figure 3.4). In those robots, encoders are therefore mainly utilized to measure speed. An accelerometer instead, by definition, measures the derivative of speed. Vision, finally, allows for the calculation of the absolute position (or the integral of speed) if the environment is equipped with unique features. An additional fundamental difference between those three sensors is the amount and kind of data they provide. An accelerometer samples real-valued quantities that are digitized with some precision. An odometer instead delivers discrete values that correspond to encoder increments. Finally, a vision sensor delivers an array of digitized real-valued quantities (namely colors). Although the information content of this sensor exceeds that of the other sensors by far, cherry-picking the information that is the most useful to complete a task remains a hard, and generally unsolved, problem.

7.1. Terminology

When dealing with sensors, it is important to provide precise definitions of terms such as "speed" and "resolution", as well as additional taxonomy that is specific to robotics. Roboticists differentiate between *active* and *passive* sensors. Active sensors emit energy of some sort and measure the reaction of the environment. Passive sensors instead measure energy from the environment. For example, most distance sensors (not including stereo vision) are active sensors, because they sense the reflection of a signal they emit; conversely, an accelerometer, a compass, or a push-button are passive sensors. Frequently the addition of an active element to a passive sensor can increase the signal-to-noise ratio of the passive sensor, so these distinctions may be blurred in some cases.

Another important term to characterize sensors is its *range*, i.e. the *difference* between the upper and the lower limit of the quantity a sensor can measure. This differs from its *dynamic range*, which is the *ratio* between the highest and lowest value a sensor can measure. It is usually expressed on a logarithmic scale (to the basis 10), also known as "decibel". The minimal distance between two values a sensor can measure is known as its *resolution*. The resolution of a sensor is primarily limited by the physical principle it leverages (e.g., a light detector can only count multiples of a quant), however it is usually limited by the analog-digital conversion process. The resolution of a sensor should not be confused with its accuracy or its precision (which are two different concepts). For example, even though an infrared distance sensor might produce 4096 different values to encode distances from 0 to $10cm$ (which suggests a resolution of around 24 micrometers), its precision is much lower than its resolution (usually in the order of millimeters) due to noise in the acquisition process.

A sensor's *accuracy* is the difference between its (average) output m and the true value v to be measured:

$$accuracy = 1 - \frac{|m - v|}{v} \tag{7.1}$$

This measure provides a quantity that approaches 1 for very accurate values and 0 if the measurements group far away from the actual value. In practice, however, this measure is rarely used and accuracy is provided with absolute values or a percentage at which a value might exceed the true measurement.

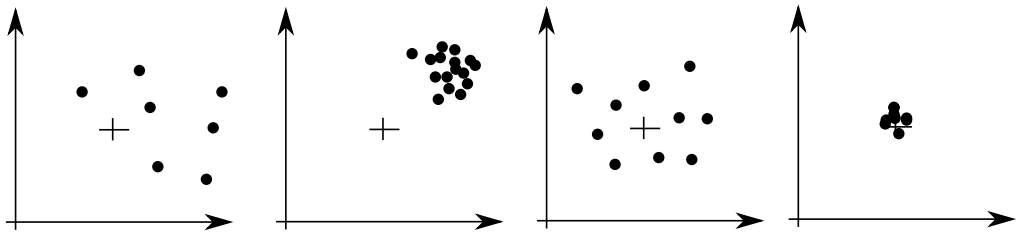

Figure 7.1. The cross corresponds to the true value of the signal. From left to right: neither precise nor accurate, precise but not accurate, accurate but not precise, accurate and precise.

A sensor's *precision* instead is given by the ratio of range and statistical variance of the signal. As detailed in Figure 7.1, precision is therefore a measure of *repeatability* of a signal, whereas accuracy describes a *systematic error* that is introduced by the sensor's physics. For example, a GPS sensor is usually precise within a few meters, but only accurate to tens of meters. This becomes most obvious when satellite configurations change, resulting in the precise region jumping by a couple of meters. In practice, this can be avoided by fusing this data with other sensors, e.g. from an IMU.

The speed at which a sensor can provide measurements is known as its *bandwidth*. For example, if a sensor has a bandwidth of 10 Hz, it will provide a signal ten times a second. This is important to know, as querying the sensor more often is a waste of computational time and potentially misleading.

7.1.1. Proprioception vs. Exteroception

Another important taxonomy is the difference between proprioception and exteroception. *Proprioception* refers to the perception of the internal state of a robot. Proprioception includes estimation of the robot's joint angles, its speeds, as well as internal torques and forces. Conversely, *exteroception* refers to sensing anything outside of the physical embodiment the robot. Exteroception is important because it is crucial for the robot to correctly perceive the state of the world, estimate the uncertainties related to it, and properly act based upon these uncertainties. Importantly, while the majority of sensor development focuses on *distal* sensors capable of measuring quantities in the far space (e.g. cameras, see Chapter 8, or sound-based sensors, see Section 7.5.3), in recent years more attention has been given to *proximal* sensors, that are concerned with measuring the environment that is immediately surrounding the body or even directly on the robot body. Applications

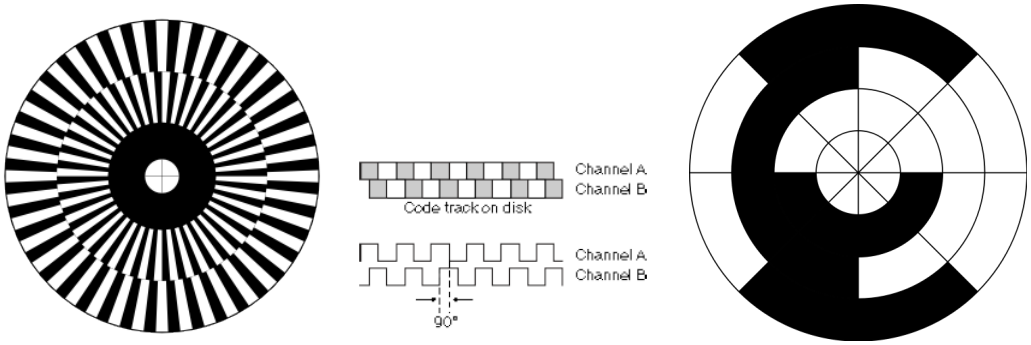

Figure 7.2. From left to right: encoder pattern used in a quadrature encoder, resulting sensor signal (forward motion), absolute encoder pattern (gray coding).

of this technology are varied, from tasks that require measuring and controlling the interaction of the end-effector with the environment (e.g. sanding a table with a fixed vertical force), to manipulating in clutter—where contact with obstacles is inevitable.

> In robotics, it often helps to make comparisons with human performance. How many daily tasks *do not* require physical interaction with the environment? If they do, would you be able to successfully complete them without contact, and how would your performance decrease if you were to "avoid collisions at all costs"?

7.2. Sensors that measure the robot's joint configuration

The most important proprioceptive sensor is the *encoder*. Encoders can be used for sensing joint position and speed, as well as force—if used in conjunction with a spring. Encoders can be divided in incremental (relative, used primarily in mobile robotics) and absolute encoders (mainly used in robot manipulators). In general, they rely on either a magnetic or optical beacon turning together with the motor and being sensed by an appropriate sensor that counts every pass-through. The most common encoder in robotics is the *quadrature encoder*, which is an optical encoder. It relies on a pattern rotating with the motor and an optical sensor that can register black/white transitions, as shown in Figure 7.2.

While a single sensor would be sufficient to detect rotational position and speed, it does not allow to determine the direction of motion. Quadrature encoders therefore have two sensors, A and B, that register an interleaving

pattern with distance of a quarter phase. If A leads B, the disk is rotating in a clockwise direction. If B leads A, then the disk is rotating in a counter-clockwise direction. It is also possible to create absolute encoders—an example of which is shown in Figure 7.2, right. This pattern is a 3-bit pattern that encodes 8 different segments on a disc. Notice that the pattern is arranged in such a way that there is only one bit changing from one segment to the other. This is known as "Gray code".

7.3. Sensors that measure ego-motion

Measuring the robot's joint configuration is limited to static observations. It does not allow the robot to detect whether it is currently moving or even accelerating (such as falling), which is particularly important for robots that are only dynamically stable such as walking humanoids or quadrotors. Motion can be estimated by relying on the principle of *inertia*. A moving mass does not lose its kinetic energy—if there is no friction. Likewise, a resting mass will resist acceleration. Both effects are due to inertia and can be exploited to measure acceleration and speed.

7.3.1. Accelerometers

An accelerometer can be thought of as a mass on a dampened spring. Considering a vertical spring with a mass attached to it, we can measure the acting force $F = kx$ (Hooke's law) by measuring the displacement x that the mass has exerted on the spring. Using the relationship $F = ma$, we can now calculate the acceleration a on the mass m. On earth, this acceleration is roughly $9.81\frac{m}{s^2}$. In practice, these spring/mass systems are realized using microelectromechanical devices (MEMS), such as a cantilevered beam whose displacement can be measured using a capacitive sensor. Accelerometers measure up to three axes of translational accelerations. Inferring an absolute position from it requires a double integration, which introduces significant noise in the estimation and makes position estimates using accelerometers infeasible in practice. However, as gravity provides a constant acceleration vector, accelerometers are very good at estimating the pose of an object with respect to gravity (i.e. roll and pitch).

7.3.2. Gyroscopes

A gyroscope is an electro-mechanical device that can measure rotational speed, and in some configurations orientation. It is complementary to the

accelerometer that measures translational acceleration. Classically, a gyroscope consists of a rotating disc that can freely rotate in a system of pivots and gimbals. When moving the system, the inertial momentum maintains the original orientation of the disc, allowing to measure the orientation of the system relative to where the system was originally. While disc-based gyroscopes are still used, for example to stabilize the cannon of a tank during motion, the mechanism remains difficult to minimize.

A variation of the gyroscope is the rate gyro, which measures rotational/angular velocities. A rate gyro can intuitively be illustrated by considering its *optical* implementation. In an optical gyro, a laser beam is split in two and sent around a circular path in two opposite directions. If the device is rotated against the direction of one of these laser beams, one laser will have to travel slightly longer than the other, leading to a measurable phase shift at the receiver. This phase shift is proportional to the *rotational speed* of the setup. As light with the same frequency and phase will add to each other and lights with the same frequency but opposite phases will cancel each other, light at the detector will be darker for high rotational velocities. Importantly, small-scale optical rate gyros are not practical, but MEMS rate gyros are widely available and use a different technology, as they rely on a mass suspended by springs. The mass is actively vibrating, making it subject to Coriolis forces when the sensor is rotated. Coriolis forces can be best understood by moving orthogonally to the direction of rotation on a vinyl disk player. In order to move in a straight line, you will not only need to move forwards, but also sideways. The necessary acceleration to change the speed of this sideways motion is counteracting the Coriolis force, which is both proportional to the lateral speed (the vibration of the mass in a MEMS sensor) and the rotational velocity, which the device wishes to measure. Note that the MEMS gyro would only be able to measure acceleration if it were not vibrating.

Rate gyroscopes can measure the rotational speed around three axes, which can be integrated to obtain absolute orientation. As an accelerometer measures along three axes of translation, the combination of both sensors can provide information on motion in all six degrees of freedom. Together with a magnetometer (compass), which provides absolute orientation, this combination is also known as *Inertial Measurement Unit* (or IMU). Note that this combination of sensors is particularly powerful, as an accelerometer and gyroscope can provide complementary information on roll and pitch, while a magnetometer and gyroscope can provide complementary information on

yaw. This innovation has powered attitude and heading reference systems (AHRS) through sensor fusion, a technique that is explored in Section Section 7.6.

7.4. Measuring force

The measurement of physical interaction forces is of paramount importance for robotics. It enables a variety of capabilities that humans take from granted, from gently picking a strawberry to safely engaging in touch-based interactions with humans.

When combining a motor and an encoder with a spring, a mechanism known as a *Series Elastic Actuator* (Pratt & Williamson 1995) , rotary and linear encoders can be used as simple force or torque sensors using Hooke's law ($F = kx$, where k is a spring constant) and x the displacement in the spring due to extension or compression. This can be used when operating a robot under a static (Section 4.1) or dynamic level of abstraction. Another method to estimate the actual force or torque acting on a joint is to measure the current consumed at each joint. Knowing a mechanism's pose allows to calculate the resulting forces and torques across the mechanism as well as the currents required for empty loading conditions. Derivations of these then correspond to additional forces that can hence be calculated.

The most accurate and most widespread device to date is the *Force/Torque (or F/T) sensor*. It is a mechanical device that is capable of detecting one or more components of a six-dimensional wrench applied to it (i.e. a 3D force and a 3D torque, see Section 4.1). Most commercially available F/T sensors use *strain gauges*. Simply put, a strain gauge is a metal (i.e. conductive) foil that changes its shape when a wrench is applied to it, and while doing so changes its electrical resistance. In a typical configuration (Figure 7.3), a F/T sensor consists of an inner hub of solid metal that is suspended in an outer ring via three symmetrical, rectangular solid metal rods. Each metal rod is equipped with a strain gauge on each side (four per rod). Typically, sensors are operated in pairs, one mounted orthogonal to the other, resulting in a total of six sensor signals, from which we can compute forces and torques in three dimensions. Such F/T sensors are available as stand-alone components that can be mounted between an end-effector and a robot arm or are integrated within the robot's joints.

While accurate and precise, F/T sensors are plagued by a number of limitations: 1) high costs due to the high precision that is required during

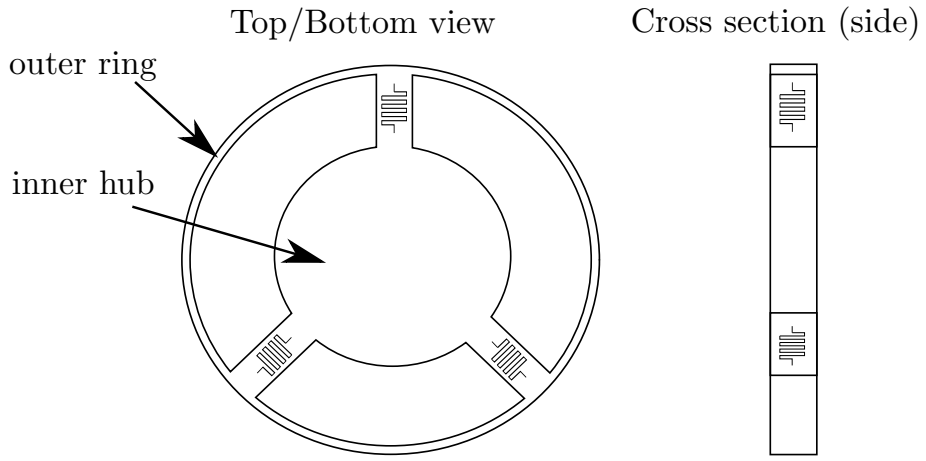

Figure 7.3. A force/torque sensor translates force and torque between two links of a robot via three metal rods that connect an inner hub to an outer ring. Here, one link of the robot connects to the outer ring, the other to the inner hub. Each metal rod is equipped with a strain gauge on each side, resulting in 12 sensors in total.

manufacturing; 2) size (a standard F/T sensor is usually the size of a human wrist); 3) low signal-to-noise ratio; 4) low bandwidth/responsivity; and 5) a single data point that is sparse in space and time. This last point becomes particularly clear when considering a robotic arm having multiple points of contact with an object. Here, a single sensor that measures forces and torques at the joint provides only very little information.

7.4.1. Measuring pressure or touch

In order to partially mitigate these limitations, roboticists have worked on a complementary capability, that is measuring the pressure applied on the robot's surface.

The human sense of touch is the oldest, the largest, and the most important of our senses. To humans, contact and physical interaction are a resource rather than an impediment, and we are surprisingly proficient at leveraging touch in a variety of situations. Therefore, it is natural for roboticists to equip robots with similar capabilities in order to achieve performance levels comparable to humans'.

A pressure sensor is a device that is capable of detecting either a contact/collision as a binary data (in which case is generally referred to as touch sensor), or a gradient of pressure applied to it. In general, the vertical

pressure applied to the sensor is proportional to the 1D vertical force that is applied to the direction normal to the sensor, and this makes a pressure sensor a good substitute to F/T sensors in specific applications (e.g. grasping). Additionally, pressure sensors are mostly based on measuring pressure through a change in *capacitance* rather than resistance (no different from the functioning of a touchscreen on a modern smartphone): when pressure is applied to a capacitor-like device (i.e. two conductive plates separated by an insulating material), the distance between the two plates reduces and this causes a change in capacitance which can be readily measured.

As introduced in the context of series-elastic actuators, distance and force sensing are tightly related via Hooke's law. There exists a large variety of touch and force sensors that rely on light-based distance measurement (see the following section) in conjunction with a flexible material with known spring constant, such as using distance sensors to measure the deformation of an elastic dome from the inside (Youssefian, Rahbar & Torres-Jara 2013) or measuring distance to objects through transparent rubber before touch and contact force after touch (Patel, Cox & Correll 2018).

If compared to F/T sensors, pressure sensors provide limited amount of information (1-dimensional vs 6-dimensional), but they allow for: 1) high responsiveness; 2) high density of sensing (up to tens of sensors per cm^2); 3) low cost; 4) ease of miniaturization.

Human touch is not limited to pressure alone, but also high-frequency information such as vibrations. These are important when discerning different surfaces. Robots can replicate this capability by measuring vibrations in the order of hundreds of Hertz by integrating accelerometers or microphones into a soft transducer and classifying spectral information (Hughes & Correll 2015).

In an extreme, it might be desirable to equip robots with an *artificial skin* that combines different sensing modalities for pressure, texture, temperature or light, possibly also including cameras or actuators to change their appearance. While there exist a variety of commercial solutions, including pressurized double-layer skin that measure pressure differentials at select locations to detect contact, as well as capacitive solutions, robotic skins have not found wide-spread applications as of yet.

7.5. Sensors to measure distance

We have seen that there is a fluent transition from proprioceptive to exteroceptive sensors as measuring the robot's internal state is tightly related to its environment as soon as contact is made. In order to explore its environment from afar, measuring distance to individual objects has shown to be critical for the robot to navigate and identify obstacles and objects of interest.

The small form factor and low price of light-sensitive semi-conductors have led to a proliferation of light-based sensors relying on a multitude of physical effects. These include reflection, phase shift, and time of flight. Other physical principles that are commonly used in distance sensors are radio (more commonly known as "Radar") and sound.

7.5.1. Reflection

Reflection is one of the easiest and most immediate principles to take advantage of: the closer an object is, the more it reflects light, radio or sound directed at it. This allows to easily measure distance to objects that reflect the signal well and that are not too far away. In order to make these sensors as independent from an object's color as possible (but unfortunately not totally independent), infrared is most commonly chosen wavelength when using light. In contrast, sound will not be effected by a surface's color, but by its surface properties and absorption characteristics.

A reflection-based distance sensor is made from two components: an emitter (that emits a signal, for example infrared light) and a receiver (tasked with measuring the strength of the reflected signal). A typical response for an infrared distance sensor is shown in Figure 7.4. The values obtained at an analog-digital converter correspond to the voltage at the infrared receiver and are saturated for low distances (flat line), and quadratically decrease afterwards.

7.5.2. Phase shift

As shown in Figure 7.4, reflection can only be precise if distances are short. Instead of measuring the strength (amplitude) of the reflected signal, laser distance sensors measure the phase difference of the reflected wave. In order to do this, the emitted light is modulated with a wave-length that exceeds the maximum distance the scanner can measure. If you were to use visible light and to do so at much slower speeds, you would see a light that keeps getting brighter, then getting darker, briefly turns off and then starts getting

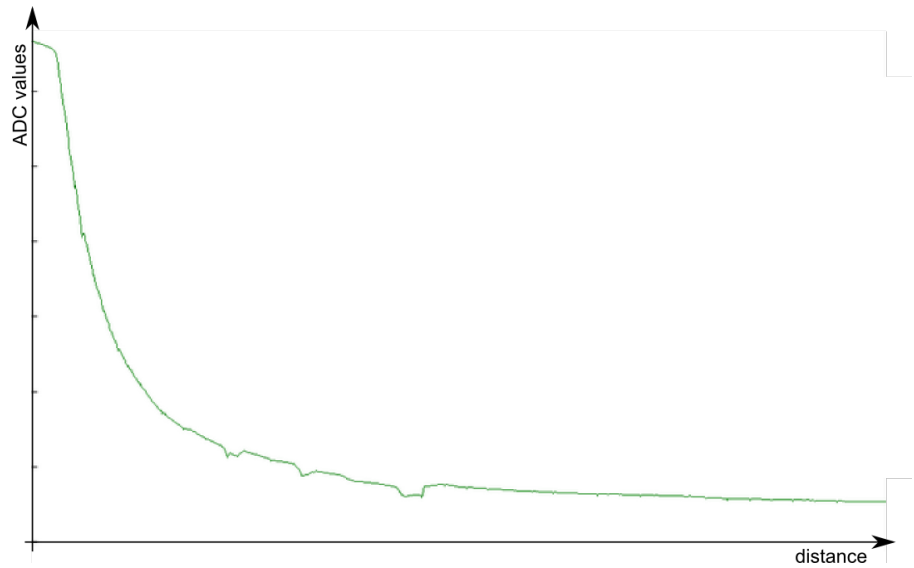

Figure 7.4. Real-world response of an infrared distance sensor as a function of distance. Units are left dimensionless intentionally.

brighter again.

Thus, if you were to plot the amplitude of the emitted signal over time (i.e., its brightness), you would see a wave that crosses zero when the light is dark. As light travels, this wave propagates through space with a constant distance (the wavelength) between its zero crossings. When it gets reflected, the same wave travels back (or at least parts of it that get scattered right back). For example, modern laser scanners emit signals with a frequency of 5 MHz (turning off 5 million times in one second). Together with the speed of light of approximately 300,000km/s, this leads to a wavelength of 60m and makes such a laser scanner useful up to 30m.

When the laser is now at a distance that corresponds exactly to one half the wave-length, the reflected signal it measures will be dark at the exact same time its emitted wave goes through a zero-crossing. Going closer to the obstacle results in an offset that can be measured. As the emitter knows the shape of the wave it emitted, it can calculate the phase difference between emitted and received signal. Knowing the wave-length it can now calculate the distance. As this process is mostly independent from ambient light, the estimates can be very precise.

As the laser distance measurement process is fast, such lasers can be combined with rotating mirrors to sweep larger areas, known as *Laser Range*

Scanners or *Lidars*. Such systems have been combined into packages consisting of up to 64 scanning lasers and are nowadays vastly used in the autonomous driving space as they are capable of providing voluminous depth data of the enviornment around a car while driving. It is also possible to modulate projected images with a phase-changing signal, which is the operational principle of early "time-of-flight" cameras, which however is not an accurate description of their operation.

7.5.3. Time-of-flight

The most precise distance measurements light can provide is by measuring its time of flight. This can be done by counting the time a signal from an emitter becomes visible in a receiver. As light travels very fast ($3 \cdot 10^8 m/s$), this requires high-speed electronics that can measure time periods smaller than nanoseconds in order to achieve centimeter accuracy. In practice, this is done by combining the receiver with a very fast electronic shutter that operates at the same frequency of the emitted light. As this timing is known, one can infer the time light has traveled by measuring the quantity of photons that made it back from the reflective surface within one shutter period. As an example, light travels $15m$ in $50ns$. Therefore, it will take a pulse of $50ns$ to return from an object at a distance of $7.5m$. If the emitter transmits a pulse of $50ns$ length and then closes the receiver with a shutter, the receiver will receive more photons the closer the object is, but no photons if the object is farther than $7.5m$. Given a fast enough and precise circuit that acts as a shutter, it is sufficient to measure the actual amount of light that returns from the emitter.

Ultrasound distance sensors

Measuring the time of flight is considerably simpler when using sound waves to measure distance (sound travels at around $344m/s$ in air). An ultrasound distance sensor operates by emitting an ultrasound pulse and measures its reflection. Unlike a light-based sensor that measures the amplitude of the reflected signal, a sound-based sensor measures the time it took for the pulse to travel back and forth. This is possible because sound travels at much lower speed ($3 \cdot 10^2 m/s$) than light ($3 \cdot 10^8 m/s$). The fact that the sensor actually has to wait for the signal to return leads to a trade-off between range and bandwidth (see Section 7.1: allowing a longer range requires waiting longer for the signal to come back, which in turn limits how often the sensor can provide a measurement. Although ultrasound distance sensors have become

progressively less common in robotics, they have an advantage over light-based sensors: instead of sending out a ray, the ultrasound pulse results in a cone with an opening angle of 20 to 40 degrees. Because of this, such sensors are able to detect small obstacles without the requirement of directly hitting them with a ray. This property makes them the sensor of choice in specific applications, such as the automated parking assist technologies in modern cars.

7.6. Sensors to sense global pose

So far, we have discussed sensors that allow the robot to measure the position of its own joints, its rotational velocity, its translational acceleration, forces from interaction with the environment, and distance to objects relative to its own pose. In order to reliably navigate in the environment, robots also need some notion of a world coordinate frame.

Localizing an object by triangulation goes back to ancient civilizations, where sailors oriented themselves using the stars. As stars are only visible during unclouded nights, seafarers have invented systems of artificial beacons emitting light, sound, and eventually radio waves. The most sophisticated of such systems is the Global Positioning System (GPS). GPS consists of a number of satellites in orbit that are equipped with knowledge about their precise location and have synchronized clocks. These satellites broadcast a radio signal that travels at the speed of light and is coded with its time of emission. GPS receivers can therefore calculate the distance to each satellite by comparing time of emission and time of arrival. As not only the position (x, y, z), but also the time difference between the GPS receiver's clock and the synchronized clocks of the satellites is unknown, four satellites are needed to obtain a "fix". Due to the way information from the satellites is coded, getting an initial fix can take on the order of minutes, but afterwards it becomes available multiple times per second. GPS measurements are neither precise nor accurate enough for robotics applications, and require to be combined with other sensors, such as IMUs. (Note that the bearing shown on some GPS receivers you may have access to is calculated from subsequent positions and is therefore meaningless if the robot is not moving.)

There exist also a variety of indoor GPS solutions, which consists of either active or passive beacons that are mounted in the environment at known locations. Passive beacons, for example infrared reflecting stickers arranged in a certain pattern or 2D barcodes, can be detected using cameras and their

pose can be calculated from their known dimensions. Active beacons instead usually emit radio, ultrasound or a combination thereof, which can then be used to estimate the robot's range to this beacon. In this domain, ultra-wideband radio in particular is common for relative localization indoors.

Take-home lessons

- Most of a robot's sensors either address the problem of determining the robot's pose or localizing and recognizing objects in its vicinity.
- Each sensor has advantages and drawbacks that are quantified in its range, precision, accuracy, and bandwidth. Therefore, robust solutions to a problem can only be achieved by combining multiple sensors with differing operation principles.
- Solid-state sensors (i.e. without mechanical parts) can be miniaturized and cheaply manufactured in quantity. This has enabled a series of affordable IMUs and 3D depth sensors that will provide the data basis for localization and object recognition on mass-market robotic systems.

Exercises

1. Given a laser scanner with an angular resolution of 0.01 rad and a maximum range of 5.6 meters, what is the minimum range d a robot needs to have from an object of 1cm width to definitely sense it, i.e., hit it with at least one of its rays? You can approximate the distance between two rays with the arc length.
2. Why does the bandwidth of a ultrasound based distance sensor decrease significantly when increasing its dynamic range, but that of a laser range scanner does not for typical operation?
3. You are designing an autonomous electric car to transport goods on campus. As you are worried about cost, you are thinking about whether to use a laser scanner or an ultrasound sensor for detecting obstacles. As you drive rather slow, you are required to sense up to 15 meters. The laser scanner you are considering can sense up to this range and has a bandwidth of 10Hz. Assume 300m/s for the speed of sound in the following.
   a) Calculate the time it takes until you hear back from the US sensor when detecting an obstacle 15m away. Assume that the robot is not moving at this point.

   b) Calculate the time it takes until you hear back from the laser scanner. Hint: you don't need the speed of light for this, the answer is in the specs above.
   c) Assume now that you are moving toward the obstacle. Which sensor will give you a measurement that is closer to your real distance at the time of reading and why?

4. Pick an educational robot platform of your choice and make a list of its sensors.

5. Construct a simple range scanner by mounting an ultrasound sensor onto a servo motor. Implement a scanning routine that allows you to collect the raw data and display it on the screen. Can you see simple features such as corners and openings?

6. Explore the internet for do-it-yourself robotics shops. What kind of sensors do they offer? What are the interfaces these sensors provide?

7. Pick a physical sensor that you have access to. Can you design an experiment to characterize its precision and accuracy?

8. Your task is to design a sensor that can detect the remaining void in a parcel for an e-commerce application.
   a) What sensors could you think off that would allow you to measure the volume of the content in the box?
   b) What additional sensors could you use assuming the box is moving on a conveyor belt.
   c) What additional information would you need to know in order to differentiate between box content, the box itself, and the environment around the box? What sensors could you use to get this information?
   d) Additional sensors are not within your budget. What kind of measures could you take to reduce the amount of information required?

9. Your task is to design an autonomous cart that can automatically dock with shelves in the environment.
   a) What kind of sensors could you use to locate the shelf in the environment? Assume that the shelf is the only object in a certain target area.
   b) What kind of physical measures could you take to simplify detection of the shelf?
   c) What kind of sensors could you use to detect whether the shelf is in a suitable position for docking?
   d) What kind of physical measures could you take to simplify the sensing process?

10. You are designing a competitive controller for the "Ratslife" game. What kind of information does the environment provide and what kind of sensor would you need to exploit it?